

6(a)	$R = \rho L / A$	C1
	$(R =) 5(.0) \times 10^{-7} \times 2(.0) / 3.3 \times 10^{-7} = 3.0 \Omega$	A1
6(b)(i)	$I = 1.2 / 3.0$	A1
	$= 0.40 \text{ A}$	
6(b)(ii)	$r = (1.50 - 1.20) / 0.40$ or $1.50 / 0.40 - 3.0$	C1
	$= 0.75 \Omega$	A1
6(c)	$E / 1.20 = 1.4 / 2.0$	C1
	$E = 0.84 \text{ V}$	A1
	or	
	$R_{XP} = (1.4 / 2.0) \times 3.0 (= 2.1 \Omega)$	(C1)
	$E = 2.1 \times 0.40$	
	$E = 0.84 \text{ V}$	(A1)
6(d)	(second wire has) larger resistance/resistance increases	M1
	p.d. across XY is larger/increases (for second wire) or p.d. across the (second) wire is larger/increases	M1
	(so) length XP (for second wire) is shorter	A1